

L1-zkEVM breakout #06

June 2026



Agenda

- Intro
- Projects updates and Q&As
- Updates from Ben

Highlights

A native zkEVM scales bandwidth, not just execution

■ Miscellaneous



It's been a while since we've seen mikeneuder — their last post was 5 months ago.



mikeneuder

2 16d

A native zkEVM scales bandwidth, not just execution



by mike neuder – sunday, june 21, 2026.

Thanks to [Ignacio](#) and [Ladislav](#) for fruitful discussions and comments.

<https://ethresear.ch/t/a-native-zkevm-scales-bandwidth-not-just-execution/25254>

zkEVM project 1 & 2

Execution Witness & Guest Program specs/test

Specs & tests

- [zkevm@v0.5.0](#)
 - Based on `glamsterdam-devnet-6`
 - Add support for external t8n filling – Using Geth to fill benchmark fixtures.
 - Spec:
 - Redefine and document SSZ list max-length in guest input.
 - Refactor and simplifications
 - Tests:
 - Malformed SSZ input, bad schema IDs, trailing bytes
 - *ChainConfig* mismatch with validated block
 - Public key validation coverage for parity mismatch in signature

Devnet guest program input exporter

- `zkevm-benchmark-workload` new CLI to automatically export `statelessInputBytes` from a joined devnet to an R2 bucket ([PR](#)).
- Allowed to find multiple bugs in Zeam and Nethermind.
- New tests were created in `forks/amsterdam`:
 - [CREATE/CREATE2 and CALL clear return data on failed pre-check](#) [Omer]
 - [Add EIP-211 return data buffer tests for SELFDESTRUCT](#) [Ignacio]

glamsterdam-devnet-5 stateless inputs

Batch-first public dataset catalog generated at 2026-07-07T13:23:24.54371136Z.

Block artifacts

1460

Batches

146

Batch size

10

Total batch bytes

1460074325

How to download

Use the batch archive links below. Each archive contains block artifacts plus a `manifest.json`.

```
curl -LO exports/batches/56370-56379.tar.zst
tar --zstd -xf 56370-56379.tar.zst
```

Verify downloads with [SHA256SUMS](#).

Machine-readable indexes

- [manifest.json](#) Dataset manifest
- [batches.jsonl](#) Batch Index

R2 public buckets do not provide directory listing; use these files instead of folder URLs.

Batch archives

Blocks	Artifacts	Size	SHA-256	Download
56370-56379	10	10720081	0x3565832b46c5d109...	56370-56379.tar.zst
56380-56389	10	10210026	0x445a85dd86deda41...	56380-56389.tar.zst
56390-56399	10	9784356	0xea610a6569649529...	56390-56399.tar.zst
56400-56409	10	10196696	0x6de4312d1557b7e8...	56400-56409.tar.zst
56410-56419	10	10542355	0xe4ed317316adc457...	56410-56419.tar.zst
56420-56429	10	10134255	0x3e71a2cd9c2ae99f...	56420-56429.tar.zst
56430-56439	10	9545769	0x639eb3f5bdfcaed7...	56430-56439.tar.zst
56440-56449	10	10970265	0x050503e4570cb8cf...	56440-56449.tar.zst

Hive-dashboard

- Unified Hive-dashboard both for:
 - *ExecutionWitness* generation spec compliance
 - Guest program test runs

The screenshot displays the 'Ethereum Hive Integration Testing' dashboard. The main section is titled 'execution-witness' and shows a summary of test results: 0 green, 2 yellow, and 1 red status icon, with 3 results and a last update time of 17 days ago. Below this, there are filters for 'Latest Test' (Clients: All, Test Suite: All) and 'Results' (Sort by Name, Sort by Coverage, Sort by Time, Group by Test, Group by Client). The dashboard lists two test suites: 'zkevm-benchmark-workload/stateless-validator' and 'eels/consume-engine-witness'. Each suite shows a list of test runs with their status (Passing or Failing), a progress bar, and a timestamp. A table at the bottom provides a detailed view of the test results.

DATE	TEST NAME	CLIENTS	TOTAL	PASS	FAIL	DIFF	STATUS
Jun 19, 2026 10:55:50	zkevm-benchmark-workload/stateless-validator	zesu_zisk-v0.18.0	23264	22638	626	—	× Failing
Jun 19, 2026 10:53:44	eels/consume-engine-witness	ethrex_rlp-engineapi	20417	20407	10	—	△ Failing
Jun 19, 2026 10:48:02	eels/consume-engine-witness	besu_rlp-engineapi	20417	20406	11	—	△ Failing

More updates

- Experimental filling using Geth for benchmark fixtures [Ignacio/Omer]
 - Successful experiment with 60M gas
 - [Fixed Geth bug in particular scenario in execution witness generation](#) [Ignacio]
- Client updates
 - Reth:
 - [Update the Reth to v2.3.0 and wire BAL checks in stateless validator](#) [Han]
 - Ethrex:
 - [~2x speedup improvement between canonical vs legacy guest program execution](#) [Han]
 - Zesu:
 - U256 Zisk accelerators experiment (~0.7% cost reduction avg case, ~4.5% worst case)

zkEVM project 6

Benchmarking and Metrics



Re-thinking Zilkworm's State and Witness management

A design discourse for improving zkEVM witness validation, and reducing overhead by directly managing state out of flat witness bytes

Updated July 3, 2026 • 19 min read •  View as Markdown



Somnath Banerjee 

[Re-thinking Zilkworm's State and Witness management](#)

ben bencik

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light 

Comparing heap allocators for zkEVM

Jun 25, 2026

[Comparing heap allocators for zkEVM](#)